

Swapnanil Mukherjee

 [linkedin.com/in/swapnanil-mukherjee](https://www.linkedin.com/in/swapnanil-mukherjee)
 github.com/SwapnanilMukherjee

 [swapnanil.mukherjee12\[at\]gmail.com](mailto:swapnanil.mukherjee12[at]gmail.com)
 [swapnanilmukherjee.github.io](https://github.com/swapnanilmukherjee)

EDUCATION

- Ashoka University** Sonepat, Haryana
• BSc. (Hons.) in Computer Science (GPA: 3.69/4) 2022 - 2026

RESEARCH INTERESTS

Multimodal Learning, Reasoning in AI, Neurosymbolic Methods, World Models, Commonsense Understanding

PUBLICATIONS/PRE-PRINTS

- Swapnanil Mukherjee**, Agyeya Negi, Tanuja Ganu, Ponnurangam Kumaraguru. (2026). C³PO: Evaluating Cross-Modal Composition and Counterfactual Performance in Omnimodal Models. *arXiv preprint arXiv:2608.05381*.
- Aritra Dutta, **Swapnanil Mukherjee**, Deepanway Ghosal, and Somak Aditya. (2025). NLKI: A Lightweight Natural Language Knowledge Integration Framework for Improving Small VLMs in Commonsense VQA Tasks. *In Findings of the Association for Computational Linguistics: EMNLP 2025, pages 10543–10563*.
- Debojyoti Chowdhury, Bhavesh Neekhara, Shreyansh Priyadarshi, **Swapnanil Mukherjee**, Debashruti Maity, Debayan Gupta, and Shubhasis Halder. (2023). Next-Gen Profiling of Tumor-resident Stem Cells using Machine Learning. *bioRxiv*.
- Swapnanil Mukherjee** and Sujit Das. (2021). Application of Transformer-Based Language Models to Detect Hate Speech in Social Media. *In Journal of Computational and Cognitive Engineering, 2(4), 278-286*.

WORK EXPERIENCE

- Microsoft Research India** Bengaluru, India
• Research Fellow July 2026 - Present
 - Working with Vineeth Balasubramaniam and Tanuja Ganu on developing neurosymbolic methods for grounded multimodal reasoning.
 - Simultaneously, working to develop a neurosymbolic multimodal verifier as a replacement for stochastic VLM judges in contexts like GRPO where deterministic verification and correctness are crucial.
- SarvamAI** Bangalore, India (Remote)
• ML Engineer Intern June - August 2024
 - Independently developed the first prototype of Sarvam's multilingual document text recognition system for 10+ Indic languages. Set up data curation pipelines, large-scale multi-node training, and experimented with different novel encoder-decoder architectures and modality fusion strategies.
 - Contributed to Sarvam's Parsing API by benchmarking, evaluating, and generating training data for the underlying models.
- Koita Centre for Digital Health - Ashoka (KCDH-A)** Sonepat, India
• Research Intern March - October 2024
 - Worked with Prof. Debayan Gupta and Prof. Steven Clipman of Johns Hopkins Medicine Institute on machine learning-based methods for enhanced forecasting of critical antiretroviral therapy (ART) drugs for the National AIDS Control Organization, Ministry of Health.
 - Set up the data acquisition and processing pipeline and experimented with various classical and neural network time-series models. Developed the first few iterations of the prediction model, and presented our initial results in a presentation to senior executives and clinicians at NACO.
 - Our method is being adopted by NACO for country-wide forecasting of 17+ HIV drugs which will optimise drug availability and impact 2.5M patients living with HIV in India.

RESEARCH EXPERIENCE

- **Modality Bias and Dominance in MLLMs:** Worked under Prof. Ponnurangam Kumaraguru Precog Lab, IIIT Hyderabad in collaboration with Tanuja Ganu to investigate modality bias in omnimodal LLMs. I led the development of **C³PO**, a benchmark that evaluates cross-modal information composition and conflict resolution across video, audio, image, and text. Our benchmark shows that current models exhibit strong modality dominance, often overlooking contradictory evidence from other modalities. (*Under review*) [**Pre-print**]
- **Undergraduate Thesis on Neurosymbolic Reasoning for VQA:** Supervised by Prof. Debayan Gupta and Prof. Somak Aditya, I extended the decompositional entailment-tree reasoning framework proposed in NELLIE and TV-TREES to enable grounded and verifiable visual question answering (VQA) on open-domain commonsense-based tasks. I also developed a *Visual Grounding Module* that grounds the proof-trees directly in token-level evidence from the image. The method surpassed comparatively baselines on challenging benchmark datasets while being fully verifiable and explainable. [**Thesis**]
- **Commonsense Reasoning in Small Vision-Language Models:** Worked under Prof. Somak Aditya (IIT Kharagpur) on the NLKI paper, which enhances small Vision-Language Models (sVLMs) for commonsense VQA tasks. By integrating externally retrieved knowledge and LLM-generated explanations, our method improved the performance of sVLMs by up to 15%, enabling them to match or outperform larger generative models. [**Paper**] [**Code**] [**Webpage**]
- **Precision Profiling of Tumor Stem Cells Using Machine Learning-Based Digital Cytometry:** Worked with Prof. Debayan Gupta and Prof Subhashis Halder's lab to develop an ML-based method for classifying and quantifying gene expression data from human stem cells, enabling highly accurate profiling of cancerous tumors to aid diagnosis and treatment. Responsible for data collation and preparation, running all ML experiments, and developing the core codebase for a software package that integrates this functionality into a user-friendly tool. Patent app. no. 202511064233. (*Under review*) [**Pre-print**]
- **Coreset Selection for Image Datasets:** Under the supervision of Prof. Raghavendra Singh, I developed a novel zero-shot method for coreset selection (on image datasets) using a simple graph-based approach for dataset distillation. The method can identify important examples without requiring any training on the target dataset, while being comparable to existing SOTA methods while being much more computationally efficient. (*Sep '23*)
- **Miscellaneous:**
 - ◊ Created an **Agentic AI system** for no-code AutoML that autonomously runs end-to-end data analysis or an ML-modelling pipeline on any dataset without user supervision, just from end-goal specifications. [**Code**]
 - ◊ Worked on a Causal Reinforcement Learning project combining Deep Q-Learning and Bayesian Inference, which taught the agent **explicit mechanics of causality** in exploration through a bayesian belief matrix and drastically improved generalization to completely unseen environments. [**Code**]
 - ◊ Developed **VeriDoc**, a protocol for verifiable integrity in public document archives. [**Code**]
 - ◊ Presented our poster on machine learning-based enhanced forecasting of antiretroviral therapy demand in India at the Johns Hopkins GKII Meet (2024) **amongst 50+ groups sponsored by the GKII Breakthrough grant** in India.
 - ◊ Core contributor on a poster titled *FedTree: Federated Learning on Tabular Medical Data* accepted and presented at the 4th Indian Cancer Genome Atlas (ICGA) Conference 2023. [**Poster**]

SKILLS AND TOOLS

- **Programming Languages:** Python, C, Java, SQL, HTML, CSS
- **Language Proficiency:** English (full professional), Bengali (native), Hindi (full professional)
- **Tools:** PyTorch, Transformers/HF, Pytorch Lightning, LlamaFactory, TransformerLens, Ollama, vLLM, CrewAI, Keras, Numpy, SKLearn, SciPy, Pandas, Matplotlib, Gradio, Git, Docker, $\text{\LaTeX}$

HONORS AND AWARDS

- Graduated *Cum Laude* from Ashoka University.
- Science Academies' Summer Research Fellowship Programme 2025 Fellow
- Dean's List (for academic excellence in a semester; **thrice**).
- College Board India Scholar 2022.